



Ahmedabad
University

CSE623 : Machine Learning

**Machine Learning-based Estimation of Muscle
Cross-Sectional Area (ACSA) from Segmented
Ultrasound Images of Lower Limb Muscles for
Sports Performance Analysis**

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Abstract

The size of muscle and the capacity to generate force is a significant measure in the analysis of sports performances, and the use of anatomical cross-sectional area (ACSA). However, ultrasound imaging is normally applied in determining the morphology of the muscles, but manual estimation of ACSA using ultrasound images is tedious and operator-based. This project investigates a classical machine learning solution to predicting ACSA of segmented ultrasound images of the Rectus Femoris (RF) and Vastus Femoris (VF) muscles. Regression models that are appropriate to small datasets are applied to interpretable geometric and shape-based features obtained out of segmentation masks. The offered solution is expected to enhance objectivity and repeatability of muscle ACSA estimation using ultrasound.

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1 What We Have Done

- Read the existing background literature about ultrasound approaches to measure muscle anatomical cross-sectional area (ACSA), in particular, the Rectus Femoris and Vastus Femoris muscles.
- Read the reference article written by Cronin et al. (2011) in order to be familiar with the automated computational analysis of skeletal muscle architecture on the basis of ultrasound images.
- Explored the extent to which the automated image-processing ideas introduced in the article can be used to decrease the use of manual muscle analysis to encourage the use of algorithmic techniques.
- Determined the applicability of classical computational and machine learning to objective extraction of muscle-related features of ultrasound images.
- Given below are some of the images of the dataset provided to us.



Figure 1: Segmentation mask of the Vastus Lateralis (VL) muscle from an ultrasound image.

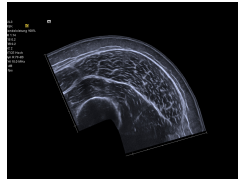


Figure 2: Original ultrasound image of the Vastus Lateralis (VL) muscle.



Figure 3: Segmentation mask of the Rectus Femoris (RF) muscle from an ultrasound image.

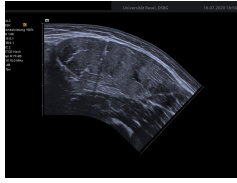


Figure 4: Original ultrasound image of the Rectus Femoris (RF) muscle.

2 What We Aspire to Do

- Arrange and authenticate the obtained segmented ultrasound images of Rectus Femoris (RF) and Vastus Femoris (VF).
- Extract quantitative features from segmentation masks, including pixel-based area, perimeter, and shape descriptors.
- Apply baseline regression models such as linear regression, support vector regression (SVR), and random forest regression to estimate anatomical cross-sectional area (ACSA).
- Evaluate model performance using appropriate statistical measures and cross-validation techniques.
- Analyze model sensitivity and interpretability with respect to segmentation quality.

References

- [1] N. J. Cronin, C. P. Carty, R. S. Barrett, and G. Lichtwark, “Automatic tracking of medial gastrocnemius fascicle length during human locomotion,” *Journal of Applied Physiology*, vol. 111, no. 5, pp. 1491–1496, 2011.
- [2] “Skeletal muscle ultrasound radiomics and machine learning for the earlier detection of type 2 diabetes mellitus,” *Journal of Medical Ultrasound*. [Online]. Available: https://journals.lww.com/jmut/fulltext/2025/04000/skeletal_muscle_ultrasound_radiomics_and_machine.5.aspx